

Software Requirements Specification

afetbilgi.com

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1 Introduction

The document is the Software Specification Requirements of the website afetbilgi.com, a website that is created by METU students and graduates to provide important information about the 6 February 2023 Pazarcık Earthquake.

1.1 Purpose of the System

The purpose of the system is to provide verified information about the 6 February 2023 Pazarcık Earthquake. The basic principles of the website are accuracy, speed, and simplicity. It is based on a human-made database. Since the earthquake is an emergency situation, accuracy and verification of information are very important and afetbilgi.com is careful to use only verified information.

1.2 Scope

The scope of the website project can be listed as follows:

- Providing important locations such as safehouses, gas stations, hospitals, and many more that are useful to victims of the earthquake.

- Providing maps that can help people locate and find places easier.
- Providing important information such as phone numbers, donation links, etc.

1.3 System Overview

1.3.1 System Perspective

AfetBilgi is not a part of large system. Afetbilgi is a simple system, and it interacts with Amazon Web Services. It uses Cloudflare to make every site you connect to the Internet secure, fast, and reliable. The basic principles of the website, which is based on a completely human-made database, are speed, accuracy, and simplicity for the simple use of all people. Since Afetbilgi is a static website, it has no large database and admins.

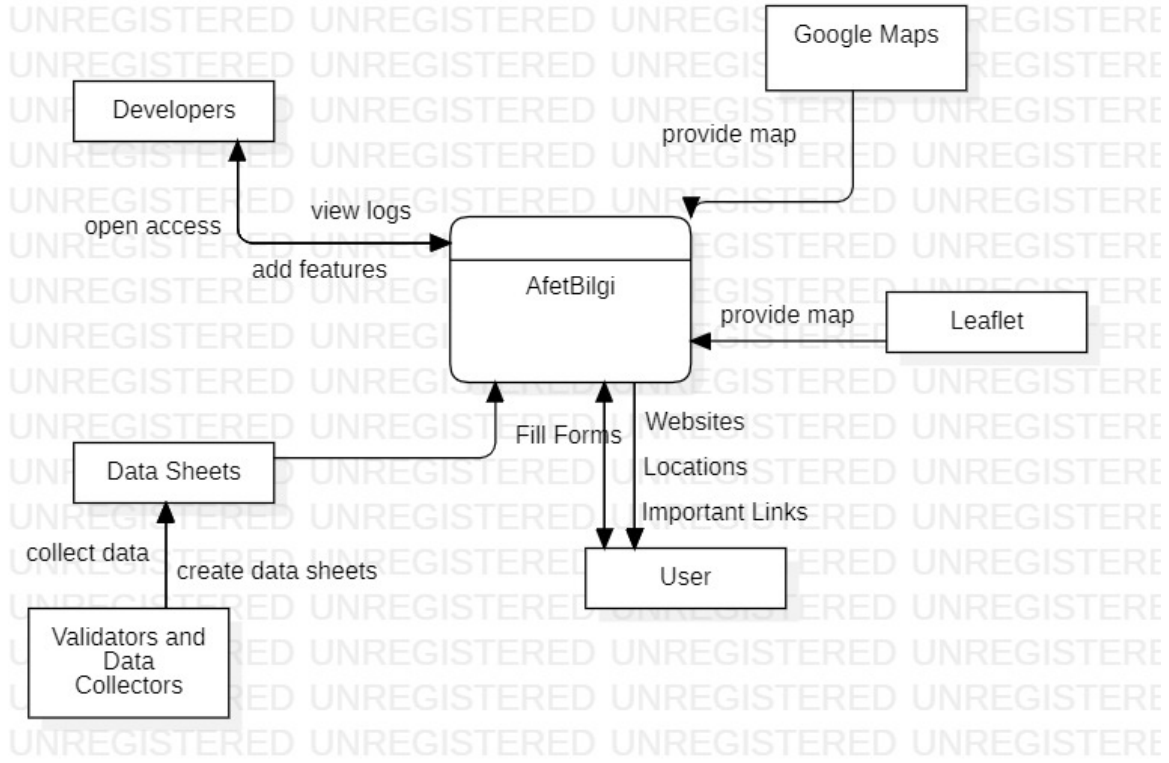


Figure 1: Context Diagram

- Developers view logs and can add features as the validators collect data after they accept the data as valid and reliable.
- Users can be provided with important links, sources, websites, locations, and related stuff. Also, they can fill out the forms on the website about a particular topic.
- Google Maps provide maps when users click to see some locations after some steps on the website.
- If the user clicks to see the map in the main interface of the system, Leaflet provides a map.

System Interfaces. Afetbilgi is a simple static website, which has no database and back-end, so It requires no server-side (also called back-end) processing, only client-side.

User Interface. The website includes End User Module as an interface.

End User Module consists of the following parts:

- Main page of the site
- Map View (for important places)
- Viewing important sources pages
- Viewing necessary places pages
- Viewing contact information page



Figure 2: Main Screen for User

[ANA SAYFA](#) [GERİ](#) Türkçe ▼

Hangi şehirde kalacak yer arıyorsunuz?

Kalacak yer imkanı sağlayan diğer kaynaklar

Otels.com
Sayfaya [bu linkten](#) ulaşabilirsiniz.

Turkish Airlines Holidays
Sayfaya [bu linkten](#) ulaşabilirsiniz.

Ahbap Güvenli Bölgeler Haritası
Sayfaya [bu linkten](#) ulaşabilirsiniz.

HepsiEmlak Dostluk Çatısı
Sayfaya [bu linkten](#) ulaşabilirsiniz.

T.C. MİLLÎ EĞİTİM BAKANLIĞI DESTEK HİZMETLERİ GENEL MÜDÜRLÜĞÜ E-DESTEK
Sayfaya [bu linkten](#) ulaşabilirsiniz.

Figure 3: Viewing Necessary Sources Screen

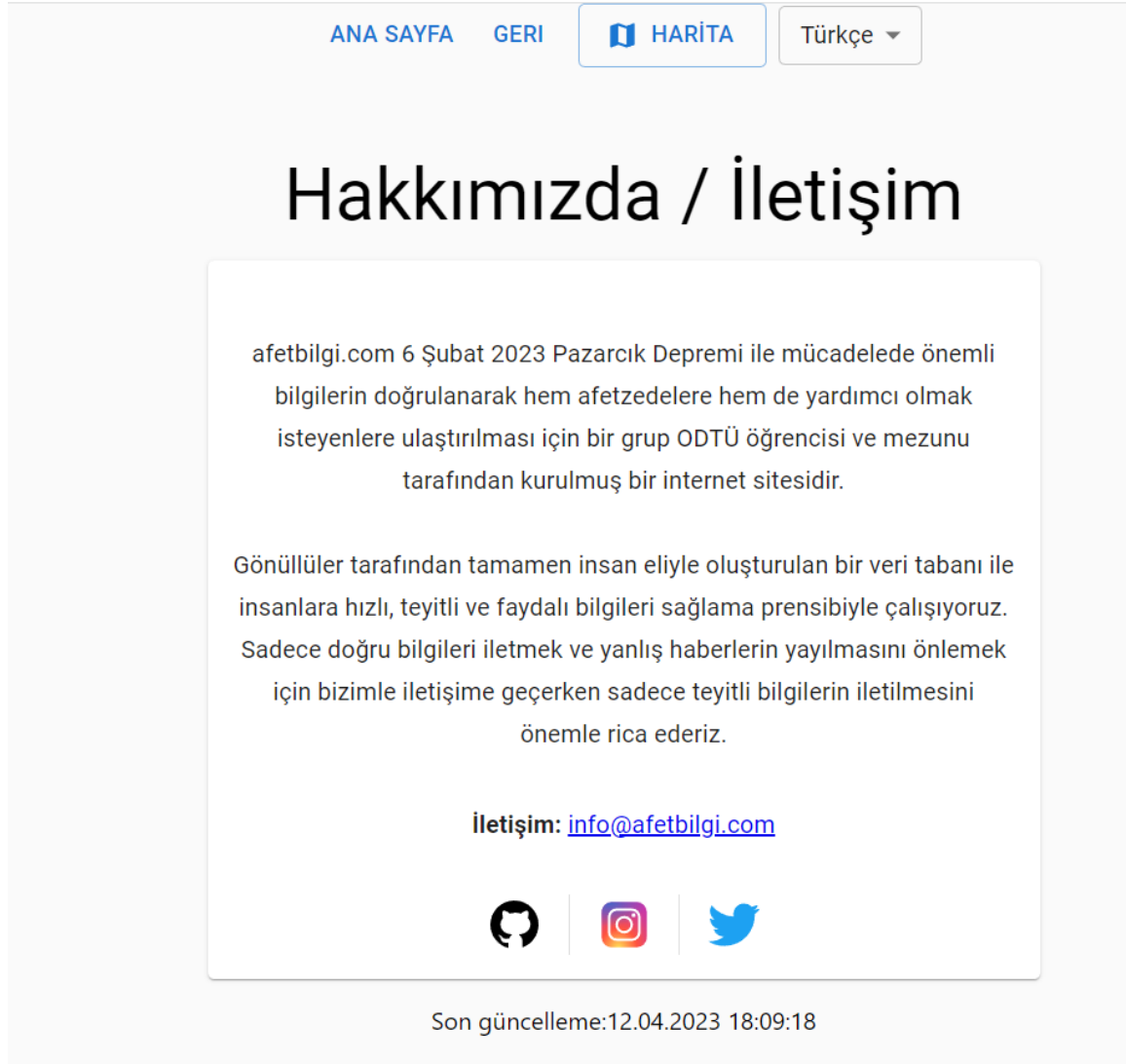


Figure 4: Viewing contact information Screen

The Screen pages above are all very simple and useful screens for the users to carry out their needs easily. End Users can find important gathering places, stations, transportation places, and hospitals in all cities of Turkey to get or provide help for earthquake victims.

Hardware Interface. The hardware interface of the system for the static website is very simple because there are no servers to maintain and the database is not large and human-made. Smartphones and computers are supported by the website.

Software Interface. Afetbilgi uses a simple human-made database to store information. For Operating Systems, different OS's, such as Windows, IOS, Android, are supported by the website. It also uses Google Maps for the locations of important places in a sub-part of the website.

Communication Interface. Afetbilgi uses HTTPS protocol for connections and security.

Memory Constraints. Memory Constraints are not a big issue for Afetbilgi system. For users, small storage of their devices is required for the system to work. RAM is the primary memory of the system while SSD is used for data storage as secondary memory.

Operations. The operations provided by Afetbilgi can be divided into:

User operations:

- Select a city
- View the map
- Change the language

- Download as pdf
- View safe assembly places
- View temporary shelters
- View active hospitals
- View Kızılay blood donation points
- View donation links
- View contact page

Worker operations:

- Collecting data
- Validating data
- Creating data sheets

1.3.2 System Functions

Function	Summary
Select a City	Users can select a city to access information about that city.
View the map	Users can open a map to see important locations on the map.
View Numbers	Users can view important numbers list.
View Donation Links	Users can view donation links as a list.
Download as PDF	Users can download all the information about the city they selected as a PDF file.
View Important Places	Users can view the important places of the city selected.

Table 1: System Functions

1.3.3 Stakeholder Characteristics

1. Developers: Developers create pages for the website using the data provided by data collectors validators. They manage everything in the system. They manage the front end, back end, etc.
2. Users: Users can access afetbilgi.com by using a web browser. They can access information that is provided to them.

1.3.4 Limitations

1. Hardware Limitations: In an emergency situation like earthquake, the website must be able to handle many users at the same time and the server needs to respond quickly.
2. Technology Requirements: The website uses Python, Typescript, Javascript, AWS, and Cloudflare therefore, they are required.
3. Parallel Operation: The website should handle many users using the website at the same time.
4. Audit functions: The project must produce accurate logs to allow auditing to work as it should.
5. Signal handshake protocols: The website is using HTTPS protocol for communication between the server and client side.
6. Safety and security: The developers are responsible to keep the website safe and secure.

1.4 Definitions

User	A person that visits the website
Developer	Manages and develops everything about the website.
Typescript	TypeScript is an open-source programming language developed and maintained by Microsoft. It is a strict syntactical superset of JavaScript, and adds optional static typing to the language. TypeScript is designed for development of large applications and transcompiles to JavaScript. As TypeScript is a superset of JavaScript, existing JavaScript programs are also valid TypeScript programs.
Python	Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level, built-in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting language to connect existing components together.
Javascript	JavaScript is a high-level, interpreted programming language that conforms to the ECMAScript specification. JavaScript has curly-bracket syntax, dynamic typing, prototype-based object-orientation, and first-class functions.
Amazon Web Services	Amazon Web Services (AWS) is a subsidiary company of Amazon, offering pay-as-you-go APIs and on-demand cloud computing services. AWS is headquartered in Seattle, Washington, and was founded in 2006. The company offers its services in eighty-four availability zones, in twenty-six geographic regions around the world. AWS's cloud infrastructure is intended to be a scalable and low-cost platform and is used in 190 countries around the world.
GitHub	GitHub, Inc. is a provider of Internet hosting for software development and version control using Git. It offers the distributed version control and source code management functionality of Git, plus its own features.
React	React (also known as React.js or ReactJS) is a JavaScript library for building user interfaces. React can be used to create single-page software applications for web and mobile platforms. The open-source library is maintained by Meta (formerly Facebook) and a community of software engineers. Although React is a library rather than a language, it is widely used in web development.

Table 2: Definitions

2 References

This document is prepared with respect to IEEE 29148 2018-11 standard: 29148 2018-11 - ISO/IEC/IEEE International Standard - Systems and software engineering - life cycle processes - requirements engineering.

References

[1] afetbilgi.com

3 Specific Requirements

3.1 External Interfaces

External Interfaces of the system includes User and Developer Interfaces. Here is the Class diagram for the External Interfaces:

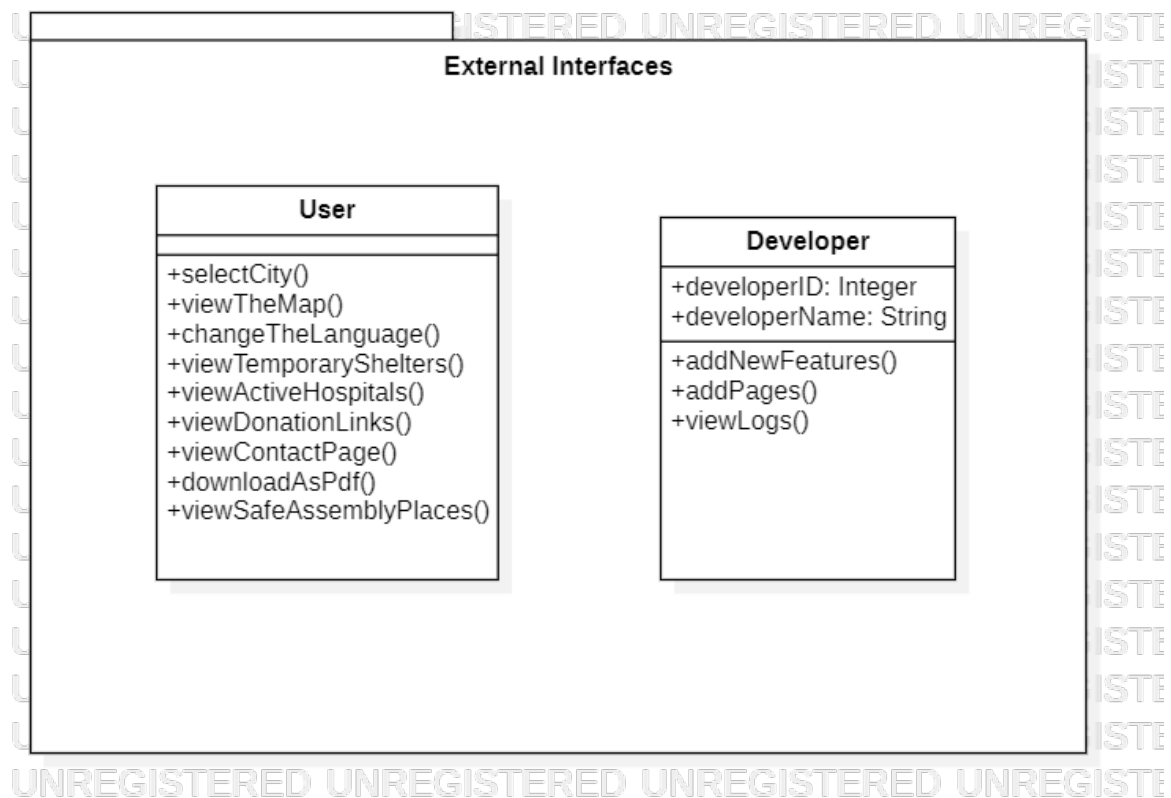


Figure 5: External Interfaces Class Diagram

3.2 Functions

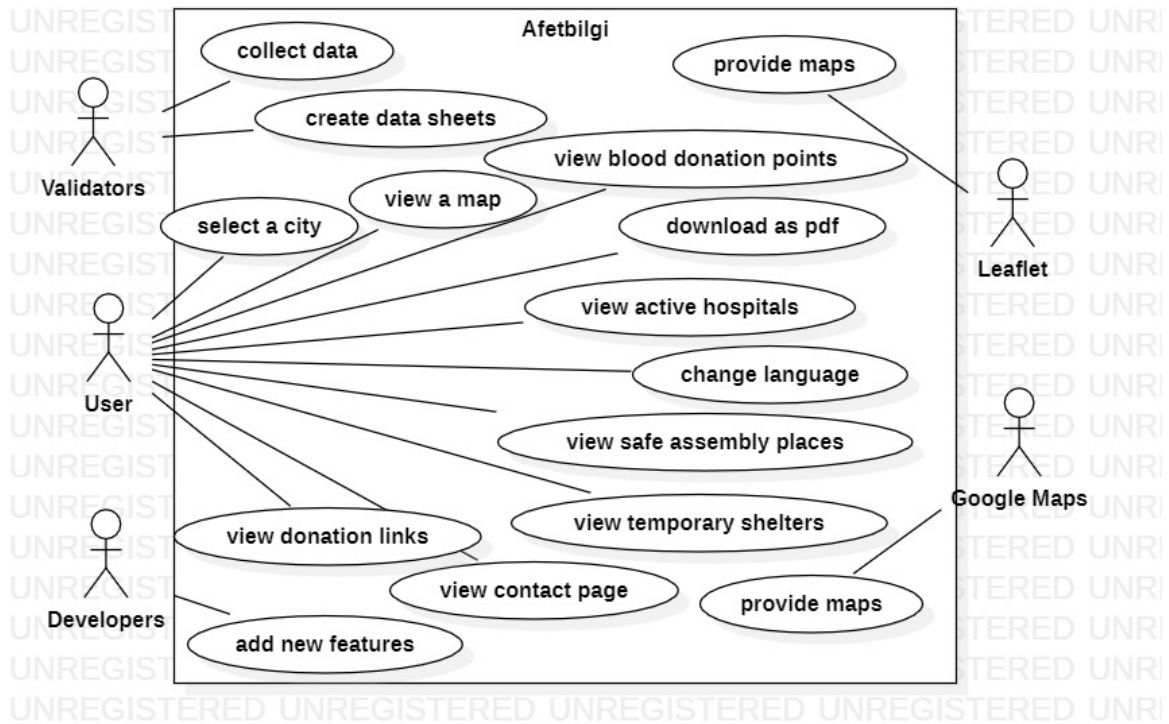


Figure 6: Use Case Model

Use-Case Name	View safe assembly places
Actors	User
Description	Users can view the safe assembly places in the cities.
Data	-
Preconditions	-
Stimulus	User pressing the view safe assembly places.
Basic Flow	Step 1 - The user clicks on the view safe assembly places button. Step 2 - The user selects a city from the menu. Step 3 - List of the safe assembly places in the city is presented to the user.
Alternative Flow#1	Step 1 - The user selects a city using the select a city button. Step 2 - The user selects a city from the menu. Step 3 - The user clicks on the view safe assembly places button. Step 4 - List of the safe assembly places in the city is presented to the user.
Alternative Flow#2	-
Exception Flow	-
Post Conditions	The user view safe assembly places.

Table 3: View safe assembly places

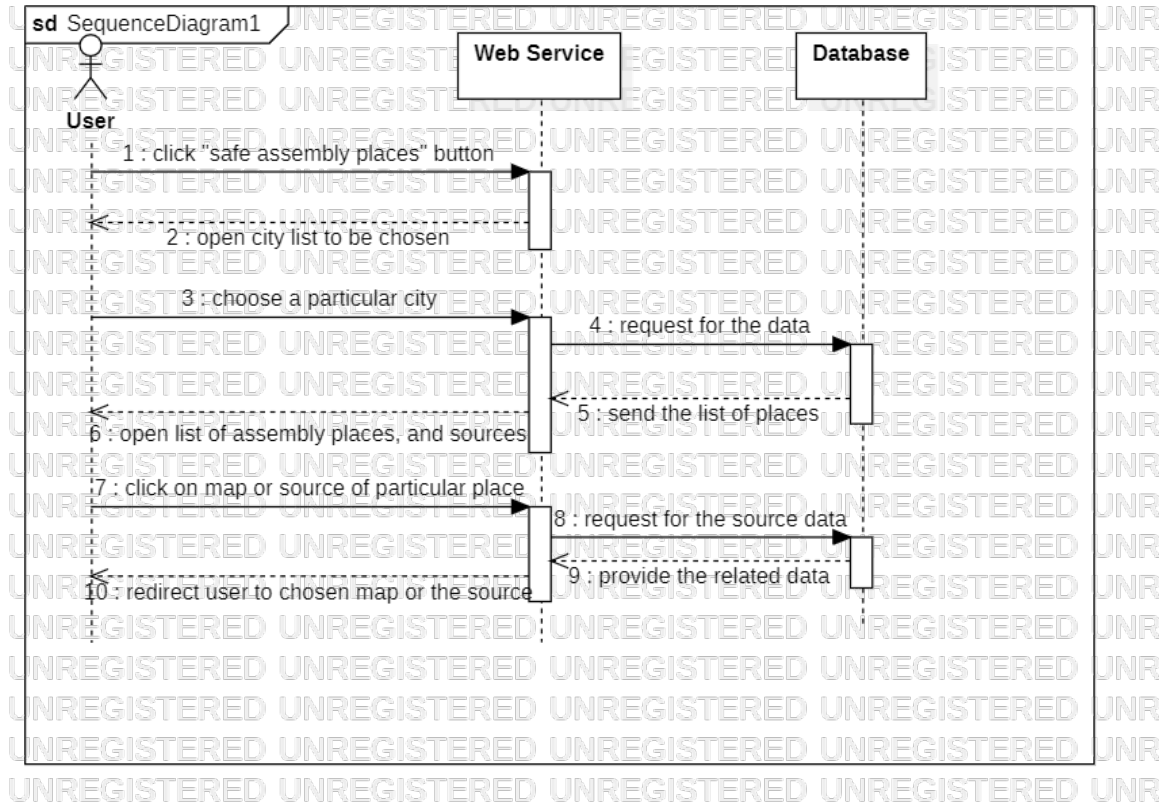


Figure 7: Sequence Diagram of "View safe assembly places"

Table 4: Collecting data

Use-Case Name	Collecting data
Actors	Validator
Description	Validators gather valuable data to donate.
Data	- Exact Places of temporary shelters
Preconditions	
Stimulus	Validators search for the data and demonstrate it.
Basic Flow	Step 1 - The validator looks for the place to give worthy information to developers.
	Step 2 - The validators decide on the reliability of the data.
	Step 3 - The validators give this worthy data to the system.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	Information is obtained by the developers.

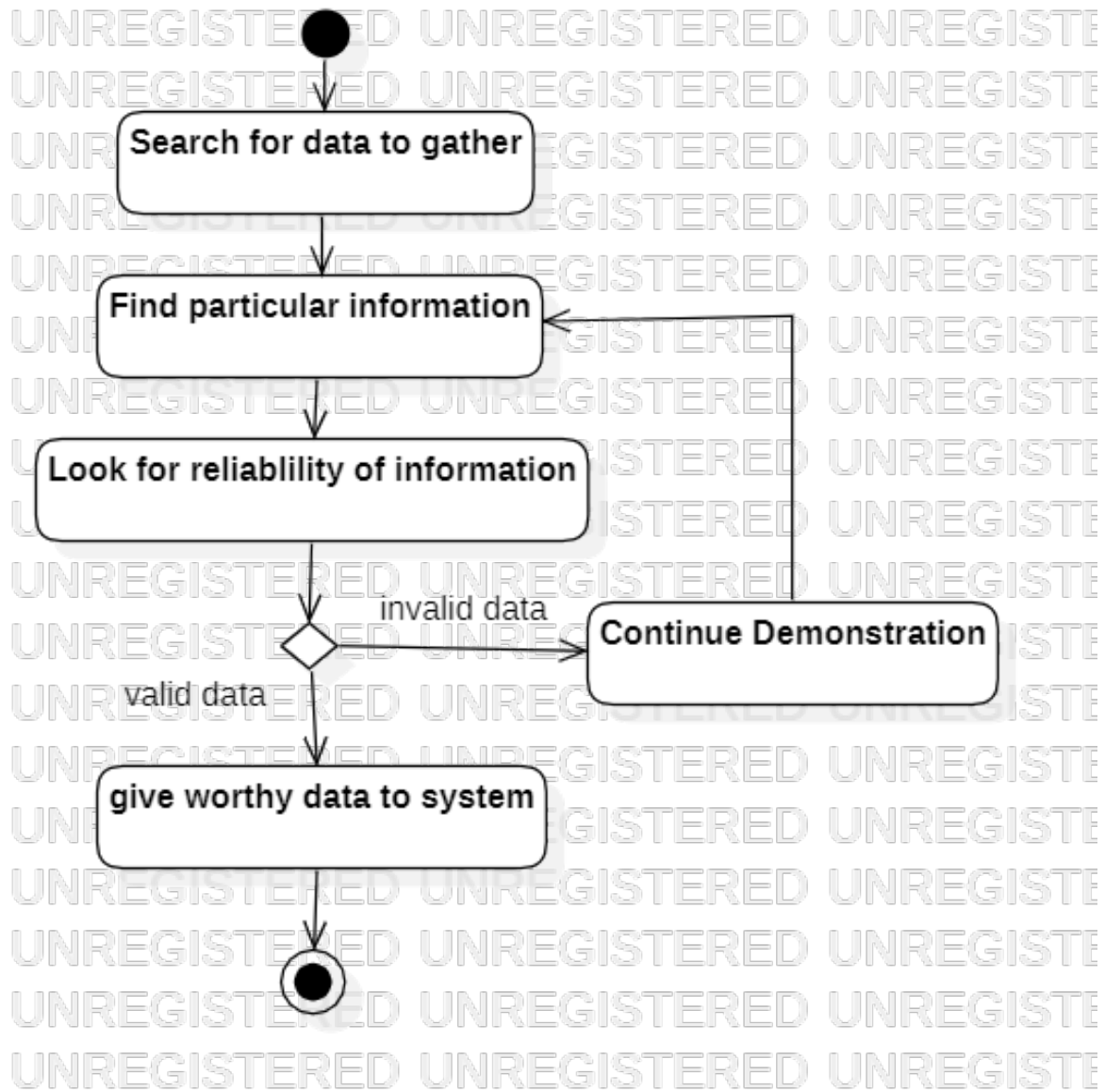


Figure 8: Activity Diagram of "Collecting Data"

Table 5: View Temporary Shelters

Use-Case Name	View Temporary Shelters
Actors	User
Description	Users can see the temporary shelters to refuge or help.
Data	- Exact Places of temporary shelters
Preconditions	
Stimulus	User pressing Temporary Shelters button.
Basic Flow	Step 1 - The user clicks on the "Temporary Shelter" button.
	Step 2 - The user clicks one of the links.
	Step 3 - The user is directed to chosen link website.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	the user is in chosen link website.

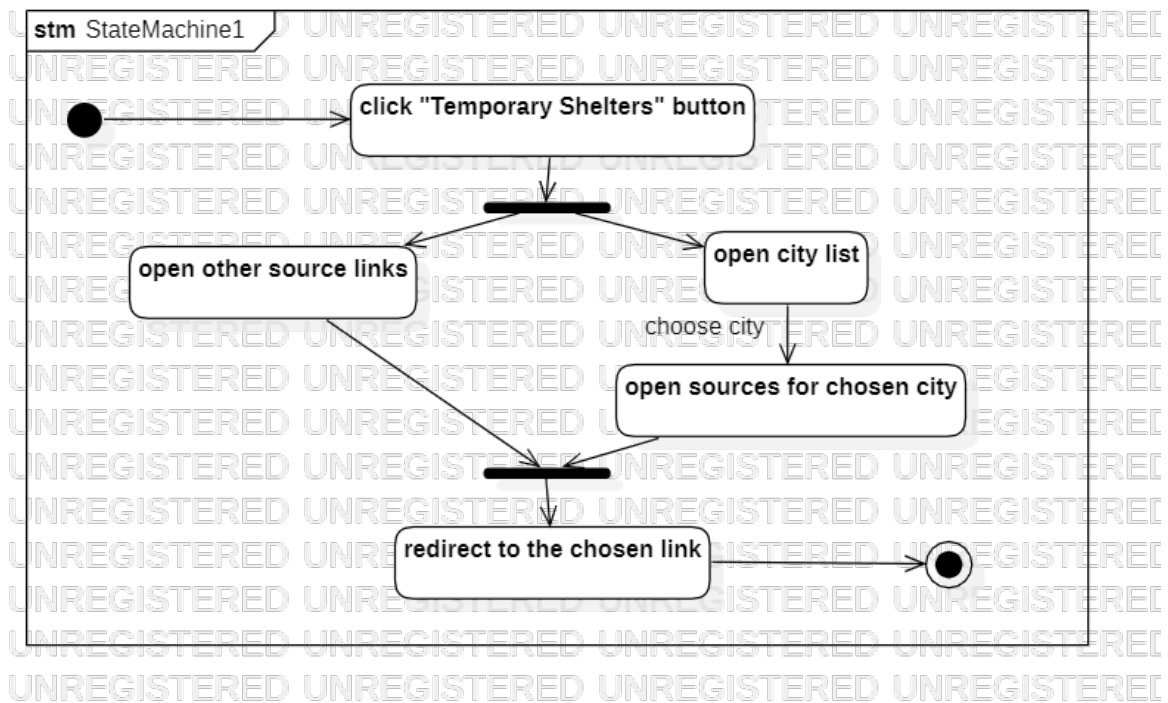


Figure 9: Statechart Diagram of "View Temporary Shelters"

Table 6: View the Map

Use-Case Name	View the Map
Actors	User
Description	Afetbilgi.com has a map where users can see important locations such as hospitals, gathering places, gas stations, etc. Users can use the map button on the page and they will be redirected to maps.afetbilgi.com for the map.
Data	Exact locations/addresses of the places shown on the map.
Preconditions	-
Stimulus	User pressing the map button
Basic Flow	Step 1 - The user clicks on the map button at the top of the page. Step 2 - The website redirects the user to maps.afetbilgi.com Step 3 - The user can interact with the map retrieved from Leaflet and the user can toggle categories from the categories tab to choose the type of locations displayed on the map.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	The user view proper places in the map .

Table 7: Select a City

Use-Case Name	Select a city
Actors	User
Description	Users can choose cities in Turkey to access the pages for the respective city.
Data	City list of Turkey
Preconditions	-
Stimulus	User selecting a city from the dropdown menu.
Basic Flow	Step 1 - The user clicks on the "Select a city" button. Step 2 - The user selects a city on the dropdown menu. Step 3 - The page is refreshed according to the city of the user. chose and the user can access information about the city chosen.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	the user read needed information about the city.

Table 8: Download as PDF

Use-Case Name	Download as PDF
Actors	User
Description	Users can download information about important locations in a city composed in a PDF file.
Data	Important locations in cities
Preconditions	-
Stimulus	User pressing the download as pdf button and choosing a city.
Basic Flow	Step 1 - The user clicks on the "PDF" button Step 2 - The user selects a city on the menu Step 3 - A pdf file of important locations about the chosen city is generated and the user is redirected to the pdf file. User can view and download the file.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	-

Use-Case Name	Change the Language
Actors	User
Description	Users can change the language of the website.
Data	-
Preconditions	-
Stimulus	User pressing the language button
Basic Flow	Step 1 - The user clicks on the language button. Step 2 - The user selects a language from the dropdown menu. Step 3 - Afetbilgi.com is now available in the chosen language.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	the user can use the system in suitable language.

Table 9: Change the language

Use-Case Name	Add new features
Actors	Developers
Description	Developers can add more pages and features to the website.
Data	Datasheets created by data collectors and validators.
Preconditions	-
Stimulus	Developers implementing the new features.
Basic Flow	Step 1 - Data comes from data sheets that are formed by data collectors and validators. Step 2 - Developers create new pages and add new features using this data.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	New features are available in the system.

Table 10: Add new features

Table 11: View Contact Page

Use-Case Name	View Contact Page
Actors	User
Description	Users can obtain contact information with developers and give worthy and reliable data about important locations.
Data	Important data in the locations
Preconditions	-
Stimulus	User pressing about/contact button.
Basic Flow	Step 1 - The user clicks on the "About/Contact" button Step 2 - The user views the contact information.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	-

Table 12: View Donation Links

Use-Case Name	View Donation Links
Actors	User
Description	Users can see the donation websites to help earthquake victims.
Data	-
Preconditions	
Stimulus	User pressing Donation Links button.
Basic Flow	Step 1 - The user clicks on the "Donation Links" button. Step 2 - The user clicks one of the links. Step 3 - The user is directed to chosen link website.
Alternative Flow#1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	the user is in chosen link website.

3.3 Usability Requirements

- A user can use all properties of the system if the Internet connection is available.
- A user can use the system without giving any personal data.
- A user can find all related sources about earthquake in only one interface.
- A user is assured by the system that all information is valid and reliable.
- The interface is in its simplest form for all users to meet possible needs
- A user can find what he or she needs in at most 4 steps.
- A user can find a particular link or source in a few minutes.

3.4 Performance Requirements

- A system should be available to many users at the same time.
- Almost 100% of the data is given as valid information by the system.
- A system is very speedy because of simple system interfaces.
- Considerable amount of data sheets can be held by the system.

3.5 Logical Database Requirements

1. The database is human-made from data collectors and validators.

2. All of the developers can access the system.
3. Location entities are filtered by cityID's to implement the select a city feature.
4. The data is retrieved as JSON files and attributes get parsed by parser scripts, then developers can use the data to display on the website.

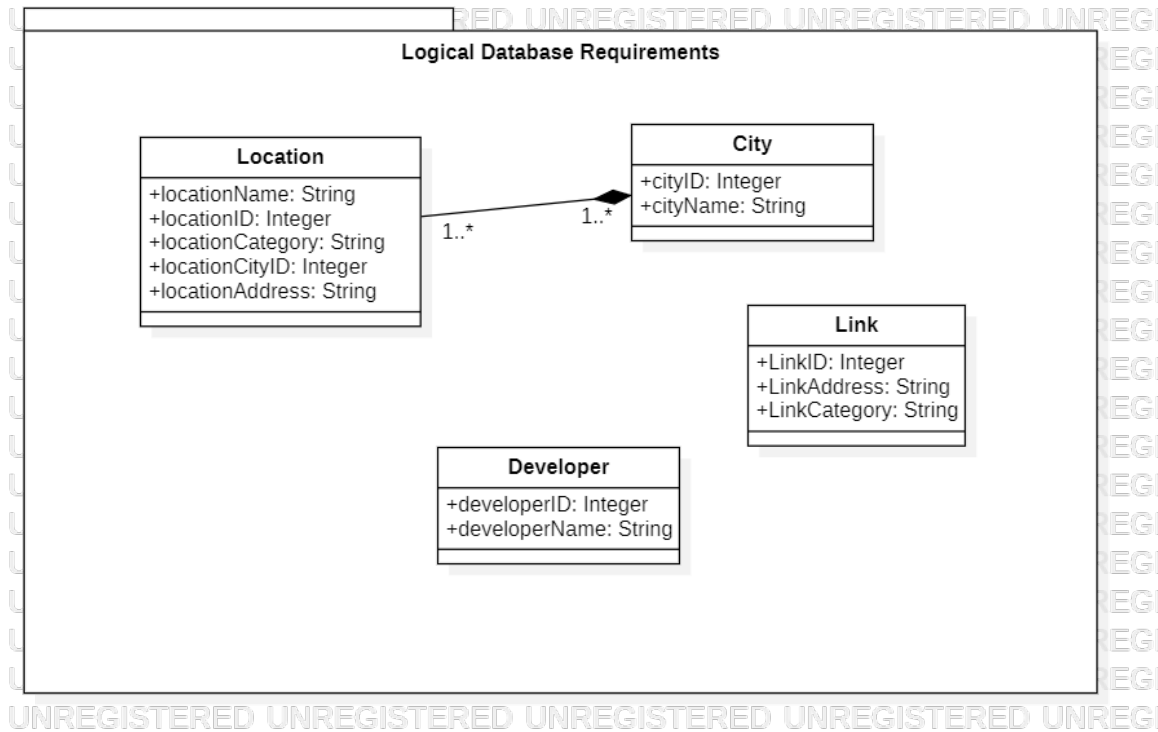


Figure 10: Logical Database Requirements Class Diagram

3.6 Design Constraints

1. The system shall be designed in accordance with the law of privacy.

2. The website shall not include any dynamic features.
3. The website shall not include any server-side scripting.

3.7 System Attributes

3.7.1 Reliability

1. The website must be durable and stable all the time in an emergency such as an earthquake.
2. The links in the website must be working correctly.
3. The sources provided by the website must be valid.

3.7.2 Availability

1. The website must be online at all times on 7/24.
2. A user can use the website as much as possible if needed.

3.7.3 Security

1. All links on the website must be secure.

3.7.4 Maintainability

1. The project must have a version control system such as Git.
2. The system must be updatable remotely.
3. Documentation of the system shall be regular and appropriate.

3.7.5 Portability

1. The website must be available on all common OS and devices such as mobile phones, computers, tablets, etc.
2. Programming language chosen for the system shall not be OS-dependable.
3. Used libraries of the system shall be applicable for different programming languages

3.7.6 Usability

1. The website must be easy to use and read. The main goal of the website is to provide fast and reliable information.
2. The website is ready to use as Internet connection is available.

3.8 Supporting Information

afetbilgi.com is a very useful source of information. It helped a lot of people in the 6 February 2023 Pazarcık Earthquake and it will be ready to help people when another natural disaster occurs in the future.

4 Suggestions to improve the existing system

This section is about improvements that can be made to afetbilgi.com, only new features are explained since the old ones are explained in the earlier sections.

4.1 System Perspective

The new feature "Get Location" is provided by Google Location Services which can retrieve users' locations from their devices. This feature can filter the unuseful information that user doesn't have to see and users can use this feature to call for help if they are in trouble from the earthquake.

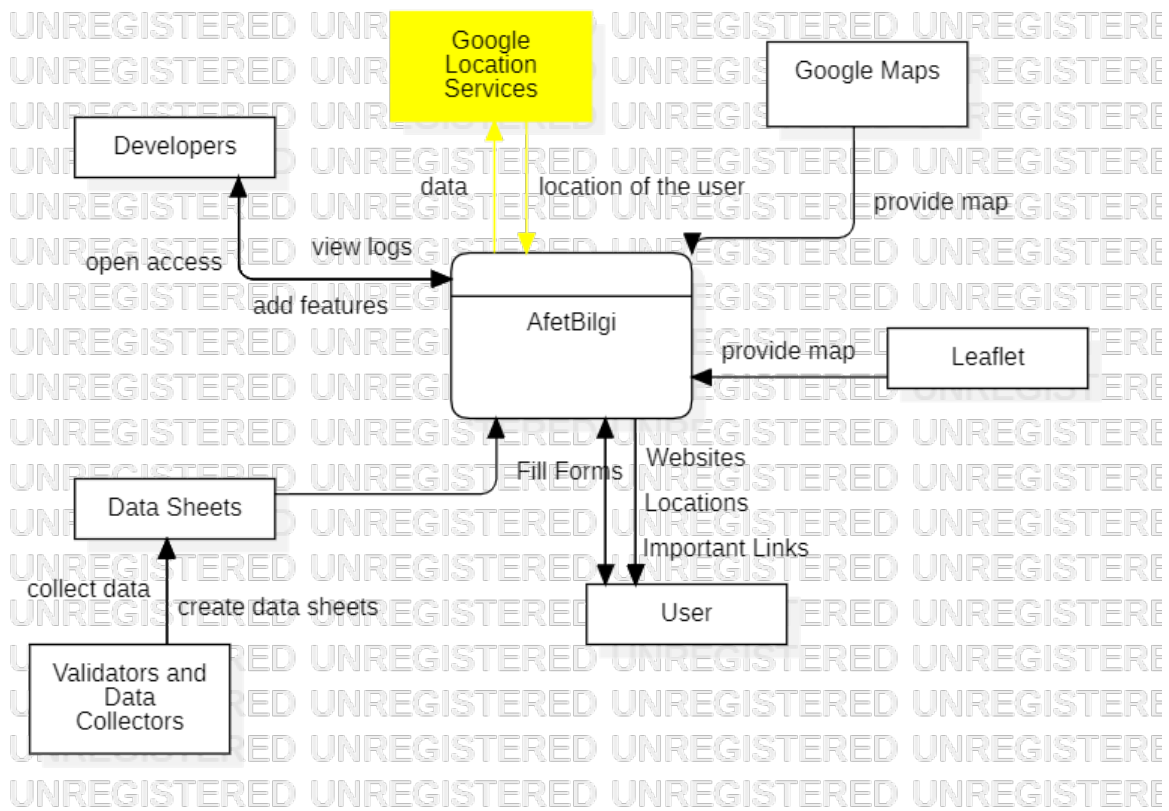


Figure 11: Context Diagram after improvements

- Google Location Services provide the location of the user immediately to the system.

- Developers view logs and can add features as the validators collect data after they accept the data as valid and reliable.
- Users can be provided with important links, sources, websites, locations, and related stuff. Also, they can fill out the forms on the website about a particular topic.
- Google Maps provide maps when users click to see some locations after some steps on the website.
- If the user clicks to see the map in the main interface of the system, Leaflet provides the map.

4.2 External Interfaces

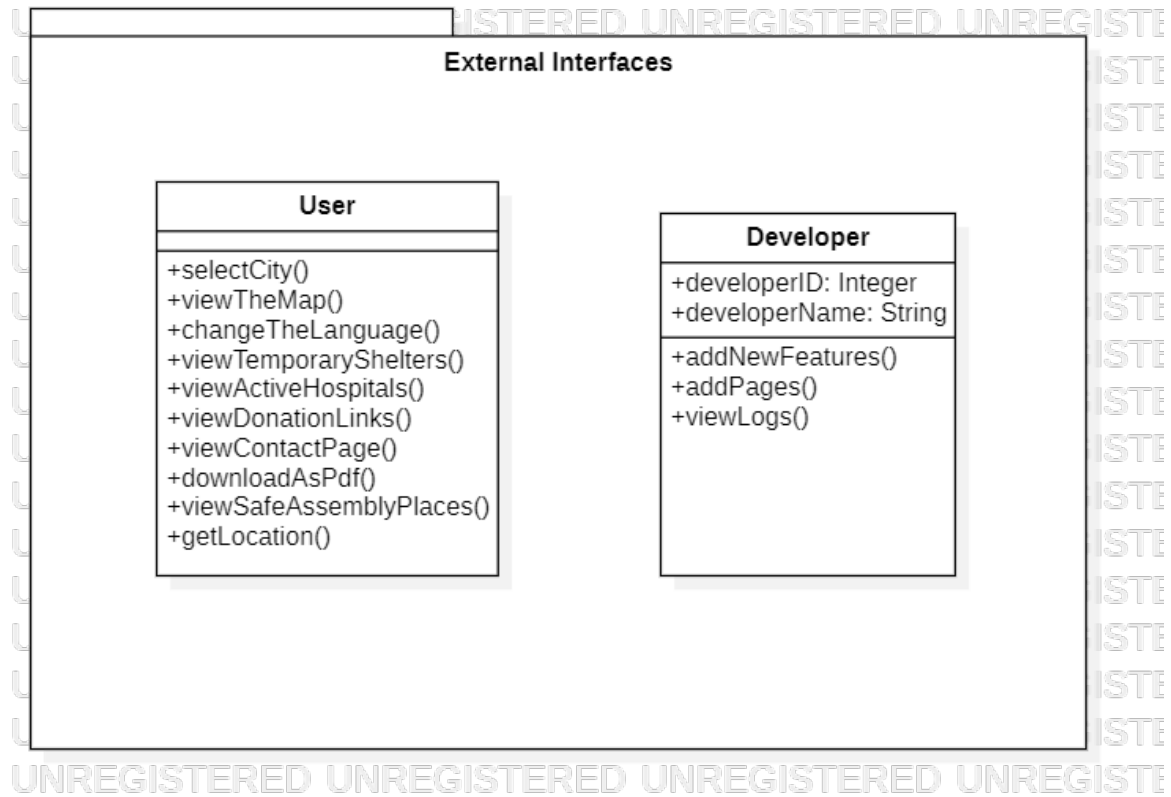


Figure 12: External Interfaces after improvements

4.3 Functions

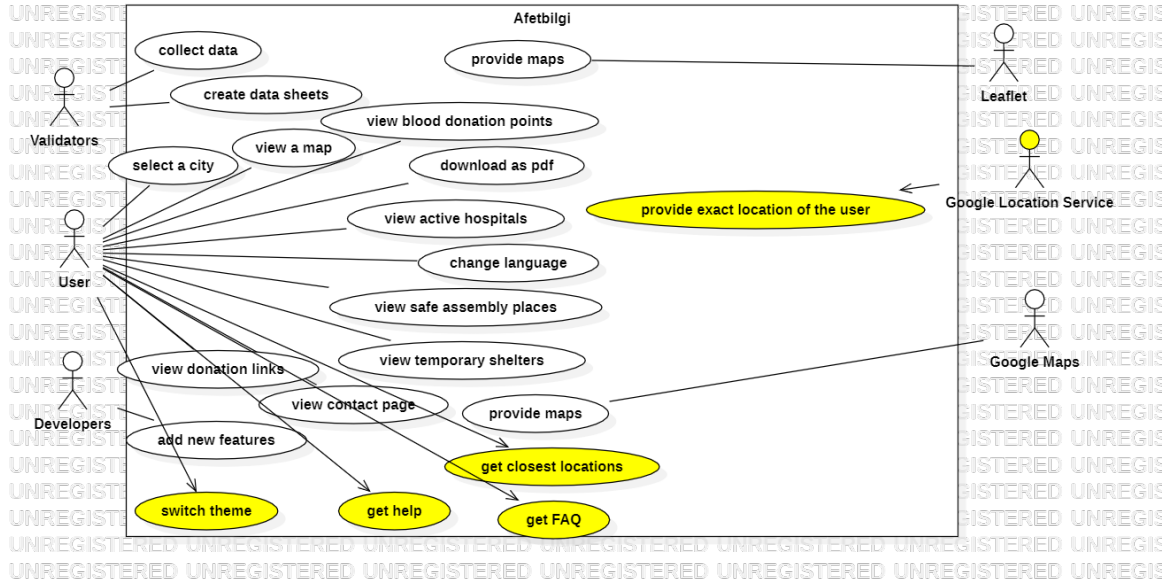


Figure 13: Use Cases after improvements

Use-Case Name	Provide Location System
Actors	User, Google Location Services
Description	afetbilgi.com can detect users location and provide information accordingly.
Data	Users location on the map
Preconditions	Users have to allow Google Location Services to locate their location from their device.
Stimulus	The user presses the get location button on the website. A pop-up is shown to the user that asks users to allow location services from their devices. After the user presses allow button, the website is refreshed according to the location.
Basic Flow	The user presses get location button. Pop-up asks if user allows their device to use google location services. User accepts and website is refreshed according to the location.
Alternative Flow #1	The user presses get location button. Pop-up asks if the user allows their device to use Google location services. User denies and uses the website without location system.
Alternative Flow#2	-
Exception Flow	-
Post Conditions	

Table 13: Provide Location System

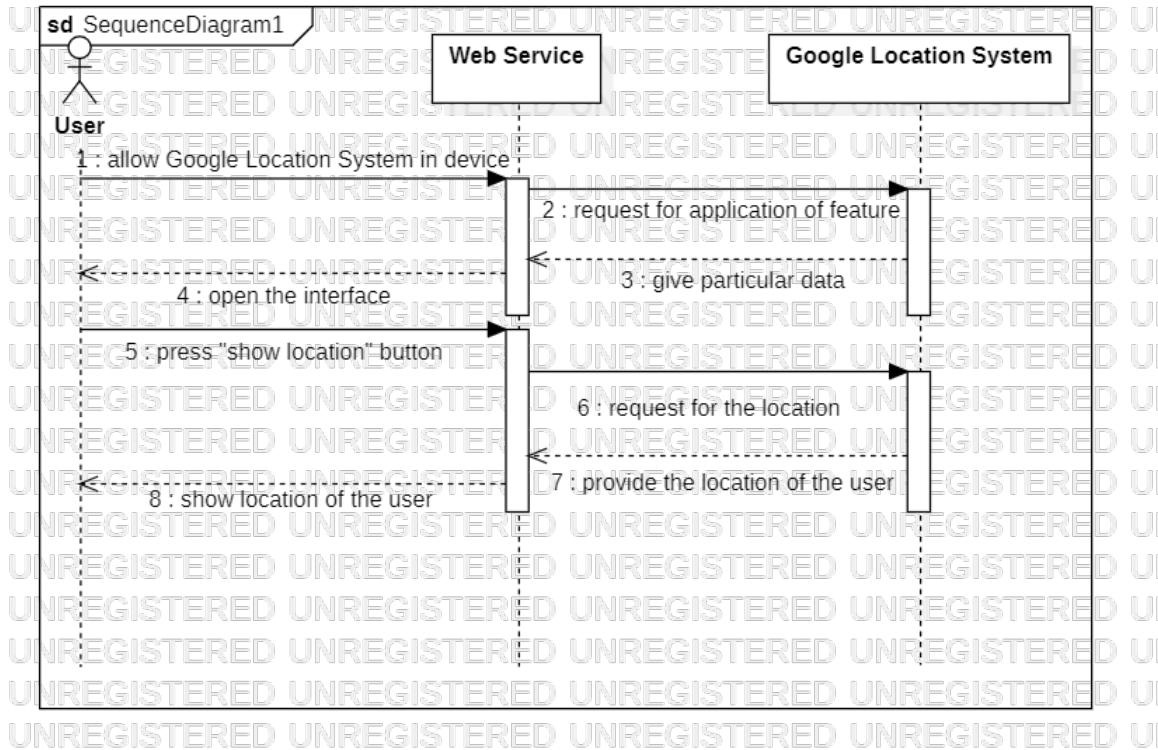


Figure 14: Sequence diagram for Provide Location System

Use-Case Name	View Closest Locations
Actors	User
Description	afetbilgi.com can provide closest locations for categories such as hospitals, restaurants, gas stations, etc. to the user.
Data	Users location from Google Location Services.
Preconditions	Users have to allow Google Location Services to locate their location from their device.
Stimulus	The user presses the get location button on the website. A pop-up is shown to the user that asks users to allow location services from their devices. After the user presses allow button, website is refreshed according to the location. Then the user can view the closes locations with "View Closest Locations" button.
Basic Flow	The user presses get location button. Pop-up asks if user allows their device to use google location services. User accepts and website is refreshed according to the location. Then user clicks to View Closest Locations button and closest locations for each category such as hospitals, restaurants, gas stations, etc. are provided.
Alternative Flow #1	The user presses get location button. Pop-up asks if the user allows their device to use Google location services. User denies and uses the website without location system. User can not use the View Closest Locations button.
Alternative Flow#2	-
Exception Flow	-
Post Conditions	

Table 14: View Closest Locations

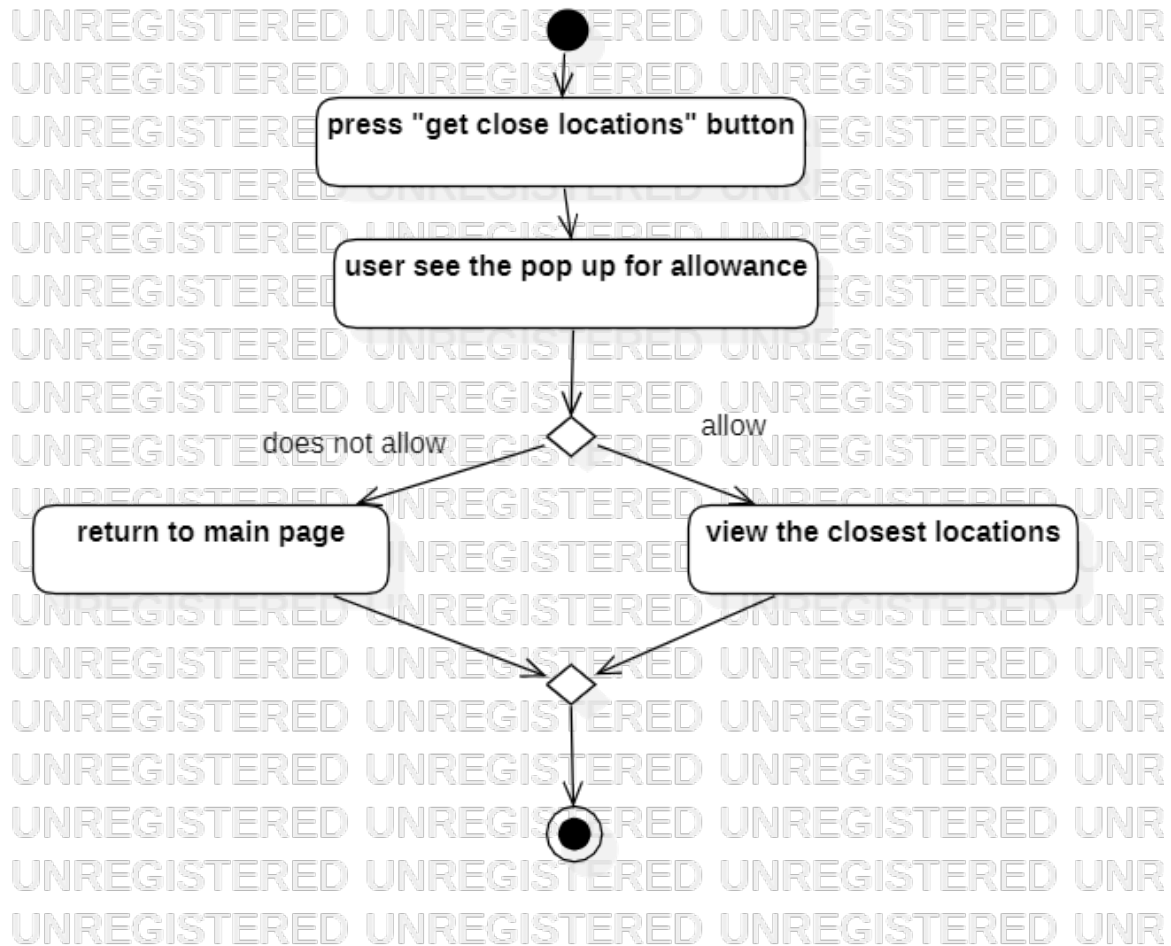


Figure 15: Activity Diagram of View Closest Locations

Use-Case Name	Call for Help
Actors	User
Description	afetbilgi.com can send data containing users' locations to organizations.
Data	Users location retrieved from Google Location Services.
Preconditions	-
Stimulus	User presses call for help button.
Basic Flow	User presses call for help button. Pop up shows and user allows location detection for their device. Location data is sent to organizations.
Alternative Flow #1	-
Alternative Flow #2	-
Exception Flow	-
Post Conditions	-

Table 15: Call for Help

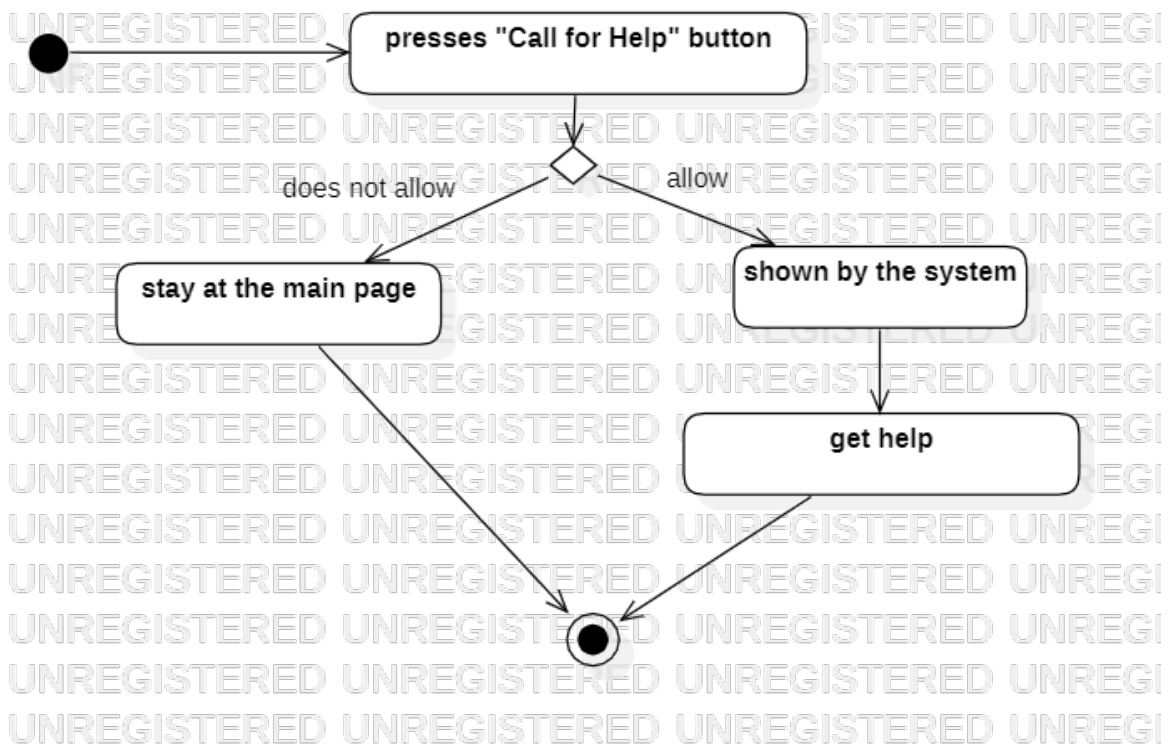


Figure 16: Statechart Diagram of Call for Help

Use-Case Name	Switch Theme
Actors	User
Description	afetbilgi.com provides a dark theme for users.
Data	-
Preconditions	-
Stimulus	User presses the dark theme button on the right top corner of the website.
Basic Flow	The user presses the dark theme button and colors of the website change to dark theme colors that prevent eyestrain.
Alternative Flow #1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	-

Table 16: Switch Theme

Use-Case Name	View Frequently Asked Questions
Actors	User
Description	Users can see faq and answers about the earthquake.
Data	-
Preconditions	-
Stimulus	User presses View FAQ
Basic Flow	User presses FAQ button. User is redirected to the FAQ page.
Alternative Flow #1	-
Alternative Flow#2	-
Exception Flow	-
Post Conditions	-

Table 17: View FAQ

4.4 Usability Requirements

For the Usability Requirements of the current system, you can refer to 3.3.

1. The user interface should have dark-mode features
2. Everything on the white theme mode should be readable on dark mode as well.

3. The system has to ask users to allow their locations from their devices.
4. The system must be able to locate users' devices correctly with Google Location Services.
5. The users' location data must be sent to the rescue organizations correctly when the user uses the Call Help feature.
6. The user must be able to change the theme using the dark theme feature using only one button click.
7. When the user uses the closest location feature, the closest locations must be calculated correctly and provided correctly to the user.
8. Access to the FAQ page must be visible enough on the page.

4.5 Performance Requirements

For the Performance Requirements of the current system, please refer to 3.4. In general, for all parts of Section 4, you can refer to the related part in Section 3 for the properties of the current system.

1. The location must be retrieved as fast as possible on the feature "Get Location".
2. The change to the dark theme must be instant.
3. Closest location features closest locations calculation must not exceed 10 seconds.
4. Call help signals must be delivered to the organizations as fast as possible.

4.6 Logical Database Requirements

All of the requirements in section 3.5 are required for the new version as well.

1. On the new version we store the locations of the users temporarily, after sending the locations to the rescue organizations, the data can be deleted if the storage is limited.

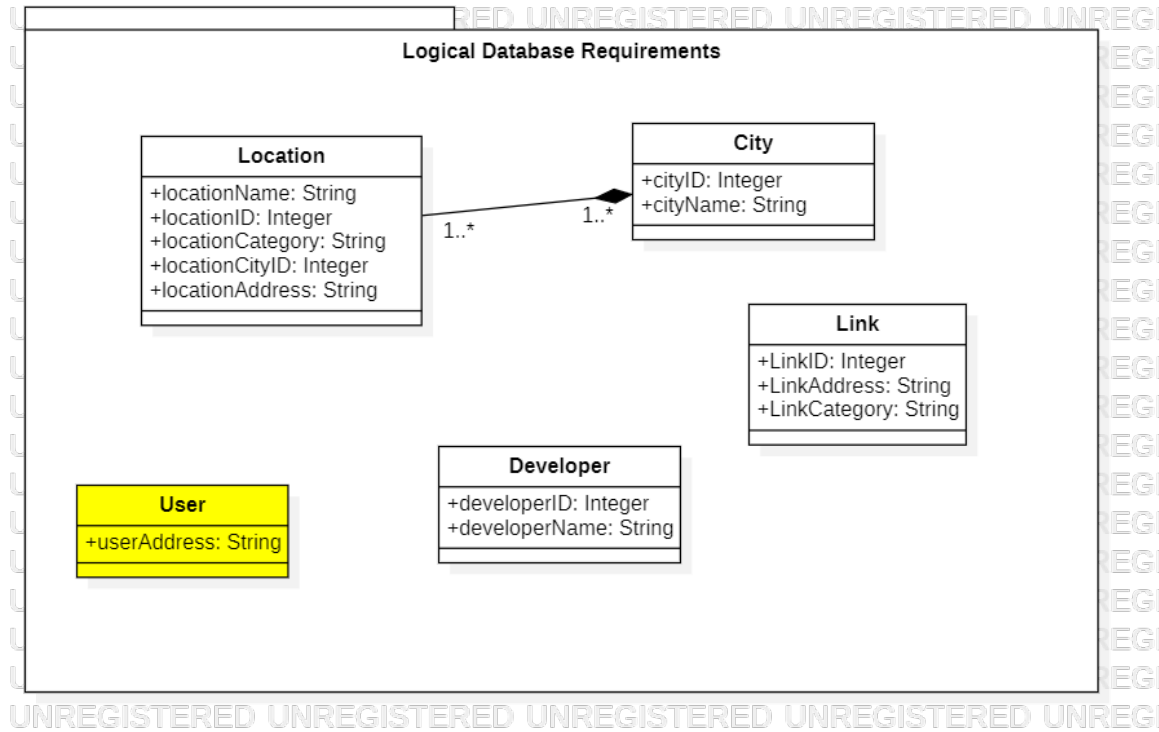


Figure 17: Database Logical Requirements Class Diagram

4.7 Design Constraints

1. All personal data of the users and all audit logs must be kept in accordance with the General Data Protection Regulation and

KVKK.

2. IP addresses, data about users' locations, and other user data are confidential, and must not be kept unless it is necessary.

4.8 System Attributes

For System Attributes, you can refer to the related part in Section 3 for properties of the current system. Here are the attributes after improvements.

4.8.1 Reliability

1. The connection with Google Location Services must be stable and reliable.
2. The dark theme must not cause any difference on the page apart from the colors.

4.8.2 Availability

1. The website must be online at all times on 7/24.
2. The new features must be ready to use at all times.

4.8.3 Security

1. Data about the locations of the users must be secured.
2. Links on the new features must be secure.

4.8.4 Maintainability

1. The new features must be updatable and bugs caused by them must be fixed in a short time.

4.8.5 Portability

1. Google Location Services must be available on all common OS and devices such as mobile phones, computers, tablets, etc.

4.8.6 Usability

1. New features must be easy to use for everyone.

4.9 Supporting Information

afetbilgi.com will be able to provide more functionality after the new improvements.